Survey Report

on the data collection and survey on the current state of water supply and drainage/sewerage in Red River Delta region

First Draft Survey Report

Vietnam Water, Sanitation and Environment Joint Stock Company (VIWASE)

Hanoi, 6/2024

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List of acronyms

ATI	Administration of Technical Infrastructure - Ministry of Construction
MARD	Ministry of Agriculture and Rural Development
BO	Build-Operation
BT	Build-Transfer
MONRE	Ministry of Natural Resources and Environment
BXD	Ministry of Construction
KEITI	Korean Environmental Industry and Technology Institute
MOF	Ministry o Finance
SR	Socialist Republic
GDP	Gross Domestic Product
PPP	Investment in the form of public-private partnership
ODA	Official Development Support
LTD	Limited
VSMT	Environmental sanitation
PC	People's Committee
O&M	Operation&Maintenance
WTP	Water Purification Treatment Plan
WWTP	Wastewater treatment plant

I. INTRODUCTION

Vietnam has 63 provinces and cities with a population of 99.2 million as of April 2022[1]. As of September 2022, Vietnam has 888 urban areas, with an urbanization rate of 41.5%[2]. The process of urbanization is taking place rapidly, the urban population is expected to increase by over 50% by 2030[3]. Therefore, it is necessary to prepare a very large infrastructure in a very short time, which includes both drainage and sanitation. Even so, the drainage and wastewater treatment system has not been fully invested and developed, the capacity of the system has not met the actual requirements and the speed of urbanization. Therefore, a part of untreated wastewater meets discharge standards into public water sources such as rivers, waterways, lakes and groundwater, causing deterioration and pollution of the water environment. Incomplete drainage system causes severe flooding in urban areas. Therefore, investment in building drainage and wastewater treatment systems has become an urgent issue in Vietnam.

In this context, on the 01st November 2023 in Hanoi, the Ministry of Environment of the Republic of Korea and the Ministry of Construction of the Socialist Republic of Vietnam has signed Memorandum of Understanding on Cooperation in urban environmental infrastructure. Based on this Memorandum of Understanding, the Korean implementing agency - the Green Transition Policy Bureau and the Vietnamese implementing agency – The Administration of Technical Infrastructure (ATI) has signed an Implementation Plan on the field of Cooperation on sharing experience on water supply and drainage/sewage.

The Korean Environmental Industry and Technology (KEITI) is required to assist in collaboration on development the overall assessment project on the current state of water supply and drainage/sewerage in Vietnam focusing on the Red River Delta region (hereinafter to referred as "KEITI Study".

In order to add more information to serve the KEITI's Study, KEITI and the Technical Infrastructure Department directed the Data survey and collection on the current status of water supply and drainage/sewerage system, focusing on urban drainage/sewerage for Red River Delta Region. The Study is researched by a consulting company, Vietnam Water and Environment Joint Stock Company (VIWASE), under the management and administration of KEITI and ATI.

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[1] https://www.qdnd.vn/y-te/tin-tuc/giua-thang-4-2023-dan-so-viet-nam-se-dat-moc-100-trieu-nguoi-21505#:~ :text=Theo%20s%E1%BB%91%20li%E1%BB%87u%20c%E1%BB%A7a%20T%E1%BB%95ng,%C4%91%E1%BA%A1t%2099%2C2%20tri%E1%BB%87u%20ng%C6%B0%E1%BB%9Di.

[2] https://moc.gov.vn/tl/tin-tuc/74077/phat-trien-do-thi-viet-nam-nhung-van-de-dat-ra-trong-giai-doan-toi.aspx

[3] https://tinhtetrunguong.vn/web/guest/thong-tin-chuyen-de/muc-tieu-den-nam-2030-ty-le-do-thi-hoa-dat-tren-50-.html

The objectives of this survey are as follows.

- To collect data on the current state of water supply and drainage/sewerage system of the Red River Delta region, focusing urban drainage/sewerage. Then conduct additional survey to gather more detail data to comprehend the current state of drainage/sewerage, focusing on urban drainage of the selected provinces (2 - 3 provinces).
- To overall assessment and select potential provinces in the Red River Delta region for urban drainage development

II. IMPLEMENTATION METHOD

KEITI and VIWASE has a pre-contract meeting on 04 April 2024 and a kick-off meeting on 03rd June 2024. The parties discussed and agreed on purpose of the study, clarify the work requirements and tentative schedule.

During the survey period, the parties shall have the meetings with Korean Expert and ATI.

The method of study will be as follow:

2.1. Desk study

Based on the available and collected documents on current water supply and drainage/sewerage system of the Red River Delta, gather relating information/data and prepare the first draft survey report.

Over past years, under ATI request, the local authorities has submitted report to ATI on the current status on the infrastructure system in the first quarter 2023 and first quarter 2024. Major data required in the KEITI report are in the 2023 report, however, some data are updated in the 2024 report are used. Some general information that are not reflect in the provinces'report, i.e.e current status of natural area, urban area, population, etc. will be used from searching the official website.

The basic data for the current status of water supply and drainage/sewerage system will include, but not limited to the following:

Basic information of water supply & wastewater treatment system, if any:

	Name of plant	Capacity (m^3/day)	Average operation flow(m^3/day)	Total population	Coverage Ratio (%)	LPCD
Water treatment plant					Supplied population	
Wastewater treatment plant					Catchment area	

The first study report shall include the excel file shall be submitted to KEITI within 15 days since the signing contract.

During the desk study, in case of necessary, the parties shall have the meetings with Korean Experts and ATI.

Upon having review/comments from the Korean experts, the parties shall discuss and select the 2-3 sites to conduct detailed data collection on site.

2.2. Data collection on site

- Based on the gathered data, KEITI experts will review and assess within 30 days to select 2-3 provinces to focus on overall assessment. The parties shall discuss, in consultation with ATI, to make a plan for site survey work to collect the needed information for the study. The meeting(s) with relevant local technical infrastructure management shall be organized through the introduction of ATI.
- The site survey works are expected to be carried out within 03 to 04 weeks.
    i. Water quality data (Influent & Effluent): at least more than 1 year
    ii. Water quality standard (regulation)
    iii. Existing plan related to the development of water supply and sewerage system.
    iv. Maps showing the location of plants, pipeline route, water supply area, catchment area, etc.
    v. Tariff structure of water supply and wastewater fees and financial report of water supply and wastewater system, if any.
    vi. Drawings and report of existing plant (if any)
    vii. Status of Intake water resources (WTP) and discharged water body (WWTP)
- The second survey report shall be submitted within 15 days after the site survey work.
- KEITI experts will review and assess the second survey report within 30 days.
- The final survey report shall be submitted within ten (15) days upon receipt of feedback from KEITI expert.
- During the study period, VIWASE experts may participate in the concerning workshop/conference as required by KEITI.

III. OUTPUT

The output shall be this report and the appendix which gather the collected information and data in the following table, including:

Table 1: General Information

Table 2: Water Supply and promulgated documents on the filed of water supply

Table 2A. Development Plan for Water Treatment Plants

Table 2B. Promulgated documents on the filed of water supply

Table 3: Drainage/Sewerage System and Promulgated documents related to the field of

drainage/sewerage

Table 3A: Wastewater Treatment Plant

Table 3B. Development Plan for Wastewater Treatment Plants

Table 3C: Promulgated documents related to the field of drainage/sewerage

Table 4: Urban drainage and drainage charge in accordance with Decree 80/2015

List of Red River Delta provinces/cities

No.	Name of cities/provinces	Grade urban
1	Hanoi City	Special
2	Haiphong City	Grade 1
3	Nam Dinh province	Grade 1
4	Ninh Bình province	Grade 2
5	Thai Binh province	Grade 2
6	Bắc Ninh province	Grade 1
7	Hưng Yên province	Grade 2
8	Hà Nam province	Grade 2
9	Hải Dương province	Grade 1
10	Vĩnh Phúc province	Grade 2
11	Quảng Ninh province	Grade 1

The urban grade is classified in accordance with Resolution No. 1210/2016/UBTVQH13 by the National Assembly Standing Committee of the Social Republic of Vietnam dated May 25, 2016 and Decision No.241/QD-TTg on approval for the national urban classification plan for period of 2021-2030 by the Prime Minister dated February 24, 2021.

Table 1-4

Name of the Red River Delta provinces/cities

No.	Name of cities/provinces	Grade urban
1	Hanoi City	Special
2	Haiphong City	Grade 1
3	Nam Dinh province	Grade 1
4	Ninh Bình province	Grade 2
5	Thai Binh province	Grade 2
6	Bắc Ninh province	Grade 1
7	Hưng Yên province	Grade 2
8	Hà Nam province	Grade 2
9	Hải Dương province	Grade 1
10	Vĩnh Phúc province	Grade 2
11	Quảng Ninh province	Grade 1

Table 1. General Information

SURVEY RESULT

Table 1. General Information

No.	Name of provinces /cities	Urban level	General information				
			Natural areas *(ha)*	Urban Areas *(ha)*	Current population *(person)*	Urban population *(person)*	Planning population to 2030 *(person)*
1	**Ha Noi**	Special	335,982	110.000-120.000	8,435,700	4,138,500	11410000 - 11959000
2	**Hai Phong**	I	152,652	152,652	2,105,026	960,640	2,300,000
3	**Hai Duong**	I	166,828	8,755	1,946,800	792,347	2,550,000
4	**Nam Dinh**	I	166,800	9,206	1897116	308,443	619,344
5	**Ninh Binh**	II	141,200	19,901	1,010,700	218,400	400,000
6	**Ha Nam**	II	86,193	16,941.10	978.100	502,551	904,800
7	**Hung Yen**	II	93,022	30,287.2	1,302,000	663,496	1,380,250
8	**Bac Ninh**	I	82,271	8187.3	1,488,300	554,600	1,800,000
9	**Thai Binh**	II	158,461	9,871	1878500	243,228	2,600,000
10	**Vinh Phuc**	II	123,600	27,844	1,300,400	450,000	1,800,000
11	**Quang Ninh**	I	620,690	204,351.0	1,398,732	916,600	2,640,000

Table 2. Water Supply System

Table 2A. Development Plan for Water Treatment Plants

Table 2B. Promulgated documents on the filed of water supply

SURVEY RESULT

Table 2. Water Supply System

No.	Name of provinces/cities	Water Supply System							
		Number of WTP *unit*	Designed capacity *m3/day*	Operating capacity *m3/day*	Pipeline system *km*	Number of P/S *unit*	Total exploitation of clean water in urban areas *m3/d*	Percentate of urban population is supplied with clean water through centralized water supply system *%*	Average of water loss, non-revenue water in Urban and Rural area *%*
1	**Ha Noi**	30	1,546,200	1,333,000				100	15
1	Luong Yen		50,000	50,000					
2	Yen Phu		100,000	80,000					
3	Ngo Si Lien		60,000	40,000					
4	Mai Dich		60,000	55,000					
5	Phap Van		30,000	8,000					
6	Ngoc Ha		30,000	25,000					
7	Ha Dinh		30,000	15,000					
8	Cao Dinh		60,000	49,000					
9	Nam Du		60,000	54,000					
10	Tuong Mai		30,000	15,000					
11	Duong Noi		30,000	25,000					
12	Ha Dong 1		20,000	22,000					

No.	Name of provinces/cities	Water Supply System							
		Number of WTP *unit*	Designed capacity *m3/day*	Operating capacity *m3/day*	Pipeline system *km*	Number of P/S *unit*	Total exploitation of clean water in urban areas *m3/d*	Percentate of urban population is supplied with clean water through centralized water supply system *%*	Average of water loss, non-revenue water in Urban and Rural area *%*
13	Ha Dong 2		22,000	22,000					
14	Small WTP (Dong My, Don Thuy)		24,000	24,000					
15	Son Tay 1		10,000	10,000					
16	Son Tay 2		20,000	20,000					
17	Small WTP (The West)		6,000	6,000					
18	Dong Anh		12,000	12,000					
19	Quang Minh Industry Area		10,000	10,000					
20	Small WTP (The North)		1,000	1,000					
21	Gia Lam		60,000	60,000					
22	Gia Lam Airport		12,000	11,000					
23	Yen Vien		7,200	7,000					
24	Small WTP (The East)		12,000	12,000					
25	Bac Thang Long		150,000	150,000					
26	Ba Vi		10,000	15,000					
27	Song Da		300,000	300,000					

No.	Name of provinces/cities	Water Supply System							
		Number of WTP *unit*	Designed capacity *m3/day*	Operating capacity *m3/day*	Pipeline system *km*	Number of P/S *unit*	Total exploitation of clean water in urban areas *m3/d*	Percentate of urban population is supplied with clean water through centralized water supply system *%*	Average of water loss, non-revenue water in Urban and Rural area *%*
28	Song Duong		300,000	220,000					
29	Water supply from Ha Nam		5,000	5,000					
30	Regional Internal surface water plants (Phuc Tho, Minh Quang, Minh Chau, Tien Thinh…)		25,000	10,000					
2	**Hai Phong**	Urban & Industrial Area: 13; Rural Area: 176	1. Hung Dao WTP: 50.000; 2. Ngu Lao: 25.000; 3. An Duong: 200.000					Urban Area: 100%; Rural Area: 100%	Urban Area: 11,7%; Rural Area: 18%
3	**Hai Duong**							Urban Area: 100%; Rural Area: 98%	Urban Area: 16-17%; Rural Area: 20%
4	**Nam Dinh**	Nam Dinh City: 01; Rural Area: 51	Nam Dinh City: 75.000					Urban Area: 86,96%; Rural Area: 96%	20
5	**Ninh Binh**	Urban and Industrial Area: 12; Rural Area: 72;	Urban and Industrial Area: 105.200; Rural Area: 53.220					Urban Area: >98%; Rural Area: 96,5%	Urban Area: <20%; Rural Area: <30%
6	**Ha Nam**	36	249,350				Urban Area: 55.700m3/day; Rural Area: 76.800m3/day; Industrial Area: 9.000m3/day	Urban Area: 97%; Rural Area: 99,215%	21.18
7	**Hung Yen**	42	129400 m3/day	99000m3/day				Urban Area: 85%; Rural Area: 78,41%	Urban Area: 20%; Rural Area: 30%
8	**Bac Ninh**	Urban Area: 12; Rural Area: 28; Industrial area: 10	Urban Area: 171400; Rural Area: 113.020; Industrial Area: 98.400	Urban Area: 113.085; Rural Area: 87.535; Industrial Area: 70.400				Urban Area: 97,5; Rural Area: 83,11	Urban Area: 20; Rural Area: 22,4
9	**Thai Binh**	Urban Area: 16; Rural Area: 59	Urban Area: 141.000; Rural Area: 231.636	Urban Area: 98.577; Rural Area: 201.630				Urban Area: 100; Rural Area: 100	Urban Area: 24,81; Rural Area: 23,76

No.	Name of provinces/cities	Water Supply System							
		Number of WTP *unit*	Designed capacity *m3/day*	Operating capacity *m3/day*	Pipeline system *km*	Number of P/S *unit*	Total exploitation of clean water in urban areas *m3/d*	Percentate of urban population is supplied with clean water through centralized water supply system *%*	Average of water loss, non-revenue water in Urban and Rural area *%*
10	**Vinh Phuc**	Total WWTs: 57 Urban area: 28 Rural area: 29	162,917	99,320				Urban area: 74% Rural area: 66,09%	Urban area: 15% Rural area: 22%
11	**Quang Ninh**								

Note: The lacking information/data shall be collected through ATI support or during the site, if selected

Table 2A. Development Plan for Water Treatment Plants

2.1. Hanoi City

No.	Water Treatment Plants (WTP)	Planning capacity of the WTP by phasing (m3/d)		
		By 2025	By 2030	By 2050
I	**Ground WTP**	**615,000**	**504,000**	**413,000**
I.1	**Central areas**	**455,000**	**393,000**	**320,000**
1	Luong Yen	40,000	40,000	35,000
2	Yen Phu	82,000	80,000	70,000
3	Ngo Sy Lien	40,000	35,000	30,000
4	Mai Dich	45,000	41,000	31,000
5	Phap Van	5,000	5,000	-
6	Ngoc Ha	25,000	21,000	16,000
7	Ha Dinh	10,000	10,000	-
8	Cao Dinh	48,000	46,000	41,000
9	Nam Du	50,000	47,000	44,000
10	Tuong Mai	15,000	5,000	-
11	Duong Noi	30,000	25,000	20,000
12	Ha Dong 1	17,000	16,000	14,000
13	Ha Dong 2	24,000	22,000	19,000
14	Other small water treatment stations (Dong My, Don Thuy, etc.)	24,000	-	-
I.2	**Western part of Hanoi city**	**32,000**	**27,000**	**20,000**
1	Son Tay 1	10,000	9,000	8,000
2	Son Tay 2	16,000	15,000	12,000
3	Small water treatment stations	6,000	3,000	-
I.3	**Norther part of Hanoi city**	**22,000**	**10,000**	**8,000**
1	Dong Anh	11,000	10,000	8,000
2	Quang Minh Industrial Zone	10,000	-	-
3	Small water treatment stations	1,000	-	-
I.4	**Eastern part of Hanoi city**	**85,000**	**74,000**	**65,000**
1	Gia Lam	55,000	55,000	50,000
2	Gia Lam Airport	11,000	10,000	8,000
3	Yen Vien	7,000	7,000	7,000
4	Small water treatment stations	12,000	2,000	-
I.5	**Southern part of Hanoi City**	**21,000**	**-**	**-**
1	Small water treatment stations	21,000	-	-
II	**Surface WTP**	**1,768,000**	**2,346,000**	**3,182,000**
	Water supply capacity for Hanoi City	**1,768,000**	**2,346,000**	**3,182,000**
	Total capacity of the WTPs	**1,918,000**	**2,571,000**	**3,582,000**
1	Thang Long North WTP	200000 - 250000	200000 - 250000	300,000
2	Ba Vi WTP	60,000	100,000	100000 - 150000
3	Da River WTP			
	Water supply capacity for Hanoi City	600,000	800,000	1,100,000
	Total capacity of the WTP	750,000	900,000	1,200,000
4	Duong River WTP			
	Water supply capacity for Hanoi City	300,000	457,000	650,000
	Total capacity of the WTP	300,000	600,000	900,000
5	Red River WTP	300,000	300,000	450,000
6	Xuan Mai WTP			
	Water supply capacity for Hanoi City	200,000	300,000	450,000
	Total capacity of the WTP			
7	Water supply from Ha Nam province (part of capacity to be provided to Hanoi)	65,000	70,000	80,000
8	Internal surface WTPs in areas (Phuc Tho, Minh Quang, Minh Chau, Tien Thinh, etc.)			
	Total water supply capacity for Hanoi City	**2,383,000**	**2,850,000**	**3,595,000**
	Total capacity of the WTPs	**2,533,000**	**3,075,000**	**3,995,000**

source : Adjustment of Water Supply Planning for Hanoi Capital to 2030, Vision to 2050
(Article 8, Decision No. 554/QĐ-TTg dated 06/04/2021 of the Prime Minister for Adjustment of Water Supply Planning for Hanoi Capital to 2030, vision to 2050)

2.2 Hai Phong City

No.	Wastewater drainage Areas	No. of proposed WTPs
I	**Maintain current capacity.**	
1	Zone 1: Central area (including Le Chan, Ngo Quyen, Hong Bang districts)	1
2	Zone 2: Central area (including Le Chan, Ngo Quyen, Hong Bang districts)	1
3	Zone 4: Western region including An Duong, An Lao Districts	1
4	Zone 5: Southern region including Tien Lang and Vinh Bao districts	1
5	Region 6: Island areas including Cat Hai and Bach Long Vi districts	4
II	**Rehabilitation, Upgrading**	
1	Zone 1: Central area (including Le Chan, Ngo Quyen, Hong Bang districts)	4
2	Zone 2: Eastern area (including Kien An - Hai An - Duong Kinh - Do Son and Kien Thuy districts)	9
3	Zone 3: Northern region including Thuy Nguyen district	10
4	Zone 4: Western region including An Duong, An Lao Districts	9
5	Zone 5: Southern region including Tien Lang and Vinh Bao districts	11
6	Region 6: Island areas including Cat Hai and Bach Long Vi districts	3
III	**New Development**	
1	Zone 2: Eastern area (including Kien An - Hai An - Duong Kinh - Do Son and Kien Thuy districts)	1
2	Zone 3: Northern region including Thuy Nguyen district	2
3	Zone 4: Western region including An Duong, An Lao Districts	2
4	Zone 5: Southern region including Tien Lang and Vinh Bao districts	2
5	Region 6: Island areas including Cat Hai and Bach Long Vi districts	1

source : Development plan for water supply, sewage networks of Hai Phong City
(Appedix VIII attached to Decision No. 1516/QĐ-TTg dated 02/12/2023 of the Prime Minister for Hai Phong provincial planning for period 2021-2030, vision to 2050)

2.3. Hai Duong province

No.	Items	Proposed location	Propose capacity (m^3/d)
I	**Major water treatment plants**		
1	ORET WTP	Hai Duong city	100,000
2	Viet Hoa WTP	Hai Duong city	100,000
3	Viwaseen 6 WTP	Hai Duong city	50,000
4	Van An WTP	Chi Linh city	40,000
5	Ben Tam WTP	Chi Linh city	10,000
6	Tu Ky WTP	Tu Ky district	35,000
7	Tuan Viet WTP	Kim Thanh district	25,000
8	Duc Chinh WTP	Cam Giang district	35,000
9	Lien Hoa WTP	Kim Thanh district	25,000
10	Cong Hoa WTP	Chi Linh city	50,000
11	Luong Dien WTP	Cam Giang district	30,000
12	Thanh Giang WTP	Thanh Mien district	30,000
13	Cam Thuong WTP	Hai Duong city	According to approved plan
II	**Other existing water treatment plant; new development projects shall be conformed to the urban planning or district region planning**		

source : Development plan for water treatment plants for Hai Duong Province

(Appedix XIII attached to Decision No. 1639/QĐ-TTg dated 19/12/2023 of the Prime Minister for Hai Duong provincial planning for period 2021-2030, vision to 2050)

2.4. Nam Dinh province

No.	Name of WTP	Proposed capacity (m^3/day)	
		Current	Period 2021-2030
I	**Upgrading**		
1	Nam Dinh City WTP	78,000	120,000
2	Yen Dinh WTP	1,000	10,000
3	Quy Nhat WTP	4,800	7,000
4	Nghia Minh WTP	7,200	7,200
5	Lieu De WTP	2,600	5,000
6	Quat Lam WTP	11,000	11,000
7	Ngo Dong WTP	2,000	5,000
8	Co Le WTP	2,000	4,000
9	Cat Thanh WTP	1,000	4,000
10	Ninh Cuong WTP	1,000	5,000
11	Xuan Truong WTP	2,000	5,000
12	Xuan Hong WTP	1,000	5,000
13	Lam Town WTP	9,000	9,000
14	Phu My Tan WTP	8,000	28,000
II	**New Development**		
1	Hai Minh WTP	-	25,000
2	Hai Trung WTP	-	25,000
3	Hai Long WTP	-	25,000
4	Rang Dong WTP	-	50,000
5	Giao Thuy WTP	-	20,000
6	Xuan Ninh WTP	-	10,000
7	Xuan Thien WTP	-	200,000
8	My Thuan Industrial area WTP	-	20,000
9	Hong Tien Industrial area WTP	-	20,000
10	Truing Thanh Industrial area WTP	-	20,000
11	Xuan Kien Industrial area WTP	-	20,000
12	Hai Long Industrial area WTP	-	50,000
13	Nam Hong Industrial area WTP	-	20,000
14	Thinh Tan Industrial area WTP	-	25,000
15	Thang Loi Industrial area WTP	-	20,000
16	Minh Chau Industrial area WTP	-	25,000
17	Lac Xuan Industrial area WTP	-	20,000
18	Thuong Thanh Industrial area WTP	-	25,000
19	Phong Hung Khanh Industrial area WTP	-	25,000
20	Giao Thinh Industrial area WTP	-	25,000
21	Nghia Son Industrial area WTP	-	25,000
22	Hai Hau Power Industrial Park WTP	-	25,000
23	Ninh Co Economic Zone WTP	-	50,000
24	Ninh Cuong Town WTP		13,000

Note: *- Proposed capacity for each phase will be based on actual usage demand and is detailed calculations in the investment project.*

- The Ninh Co Economic Zone Water Supply Plant will be implemented according to the overall construction Plan of the Economic Zone.

source : Development plan for water supply plants for Urban, Industrial for Nam Dinh Province
(Appedix X attached to Decision No. 1729/QĐ-TTg dated 29/12/2023 of the Prime Minister for Nam Dinh provincial planning for period 2021-2030)

2.5. Ninh Binh province

No.	Name of WTP	Proposed capacity (m^3/day)	Proposed location and Project Scope
A	**URBAN WATER SUPPLY**		
II	**Rehabilitation and Upgrading of Existing Water Treatment Plant**		
1	Tam Diep WTP (period 2021-2030)	12.000 to 25.000	Tam Diep city and related localities
3	Hoang Long WTP (period 2021-2030)	20.000 to 40.000	Gia Vien District and related localities
II	**New construction water treatment plants**		
5	Yen Dong WTP (period 2021-2030)	40,000	Yen Mo District and related localities
6	Nho Quan 2 WTP (period 2021-2030)	35,000	Nho Quan District and related localities
7	Yen Dong WTP (period 2031-2050)	40.000 to 80.000	Yen Mo District and related localities
8	Nho Quan 2 WTP (period 2031-2050)	35.000 to 70.000	Nho Quan District and related localities
B	**RURAL WATER SUPPLY**		
I	**Rehabilitation and Upgrading of Existing Water Treatment Plant**		
1	Rural water supply works (period 2021-2030)	According to demand and supply capability	Disgtricts, cities
II	**New construction water treatment plants**		
1	Binh Minh WTP (period 2021-2030)	6,000	Kim Son District
2	Xuan Chinh WTP (period 2021-2030)	5,000	Kim Son District
3	Ninh Van WTP		Hoa Lu District

Note : Other rural and urban water supply works are invested in new construction, renovation and upgrading to ensure that the rate of rural households using hygienic domestic water reaches 100%, of which 85% is used clean water

source : Development plan for Irrigation, Dike Systems, and Water Supply Networks for Ninh Binh Province (Appedix II attached to Decision No. 218/QĐ-TTg dated 04/03/2024 of the Prime Minister for Ninh Binh provincial planning for period 2021-2030, vision to 2050)

2.6. Ha Nam province

No.	Content	Proposed capacity (m^3/day)	Development Orientation
A	**Rehabilitation and Upgrading water supply plant**		
1	Water treatment plants on the Red River	600,000	Renovation, upgrading capacity and new construction
2	Water treatment plants on the Day River	200,000	Renovation, upgrading capacity and new construction
B	**PRIMARY PIPELINE NETWORK**		Construction of transmission pipeline systems along main traffic axes connecting water treatment plants and water supply areas to ensure water supply safety
C	**Investment in repairing, upgrading, and constructing new centralized household water supply facilities within the province.**		

source : Development plan for water treatment plants for Ha Nam Province
(Appedix XII attached to Decision No. 1686/QĐ-TTg dated 26/12/2023 of the Prime Minister for Ha Nam provincial planning for period 2021-2030, vision to 2050)

2.7. Hung Yen province

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2.8. Bac Ninh province

No.	Name of WTP	Proposed capacity (m^3/day)	Proposed location
A	**URBAN WATER SUPPLY**		
I	**Rehabilitation and Upgrading of Existing Water Treatment Plant**		
	Northern area of Duong River		
1	Bac Ninh WTP(upgrading capacity)	160,000	Que Vo Town
2	Tri Phuong WTP(upgrading capacity)	50,000	Tien Du District
3	Tam Giang WTP(upgrading capacity)	40,000	Yen Phong District
4	Expansion of Yen Phong 1 WTP	40,000	Yen Phong District
5	VSIP Industrial Park WTP	23,000	Tu Son City
6	Song Cau WTP	15,000	Bac Ninh City
7	Duc Long WTP	12,000	Que Vo Town
	South area of Duong River		
1	Dinh To WTP(upgrading capacity)	40,000	Thuan Thanh Town
2	Quang Phu WTP(upgrading capacity)	20,000	Luong Tai District
3	An Thinh WTP	20,000	Luong Tai District
4	Dai Lai centralized clean water plant (upgrading capacity)	15,000	Gia Binh District
5	Thuan Thanh III WTP	15,000	Thuan Thanh Town
6	Centralized clean water supply works in Song Ho ward, Thuan Thanh town	15,000	Thuan Thanh Town
7	Gia Binh centralized clean water supply works	10,000	Gia Binh District
8	Dai Bai WTP	10,000	Gia Binh District
9	Phu Hoa commune centralized clean water supply, Luong Tai District	10,000	Luong Tai District
10	Tan Chi WTP	10,000	Tien Du District
B	**New construction water treatment plants**		
	Northern area of Duong River		
1	Yen Phong 2 WTP	5,000	Yen Phong District
2	Nam Son Industrial Park WTP	25,000	Bac Ninh City
3	Que Vo III Industrial Park WTP	16,000	Que Vo Town
	South area of Duong River		
1	Khai Son Industrial park WTP	5,000	Thuan Thanh Town
2	Gia Binh Industrial Park WTP	25,000	Gia Binh District
3	Gia Binh II Industrial Park WTP	20,000	Gia Binh District
4	Gia Binh I Industrial Park WTP	5,000	Gia Binh District
5	Thuan Thanh II Industrial Park WTP	4,000	Thuan Thanh Town
6	Thua Town WTP	3,000	Luong Tai District
B.	**RURAL WATER SUPPLY**		

Note : Rural water supply works are invested in new construction, renovation and upgrading to ensure that the rate of rural households using hygienic domestic water reaches 100%, of which 100% is used clean water

source : Development plan for water treatment plants for Bac Ninh Province (Appedix XII attached to Decision No. 1589/QĐ-TTg dated 08 /12/2023 of the Prime Minister for Bac Ninh provincial planning for period 2021-2030, vision to 2050)

2.9. Thai Binh province

No.	Name of WTP	Proposed capacity (m^3/day)	No.	Name of WTP	Proposed capacity (m^3/day)
I	**Thai Binh City**		38	Thuy Quynh Commune WTP	1,500
1	Thai Binh City WTP	70,000	39	Thai Thuy Commune WTP	2,000
2	Hoang Dieu WTP	40,000	40	Thuy Dung Commune WTP	5,500
3	Nam Long No.1 WTP	30,000	41	Thai Hung Commune WTP	2,500
4	Nam Long No.2 WTP	20,000	42	My Loc Commune WTP	5,000
5	Dong Tho WTP (New construction)	50,000	43	Diem Dien Town WTP	60,000
II	**Huyện Đông Hưng**		44	Amonitrat Factory WTP	12,000
6	Bach Dang Commune WTP	23,000	**VI**	**Tien Hai District**	
7	Dong Dong WTp	14,000	45	Nam Chinh Commune WTP	80,000
8	Dong Huy Commune WTP	10,500	46	Dong Trung Commune WTP	45,000
9	Phong Chau Commune WTP	14,000	47	Tam Dong, Vu Lang Commune WTP	1,500
10	Dong Cuong Commune WTP	7,000	48	Dong Minh Commune WTP	10,500
11	Trong Quan Commune WTP	5,000	49	Tay Tien Commune WTP	1,500
12	Dong Giang Commune WTP	1,000	50	Vu Lang Commune WTP	1,500
13	Dong Hung Commune WTP	12,000	51	Van Truong Commune WTP	1,000
14	Thang Long WTP	2,500	52	Dong Tra Commune WTP	1,000
III	**Hung Ha District**		53	Nam Cuong Commune WTP	1,000
15	Minh Tan Commune WTP	25,000	54	Tien Hai Town	50,000
16	Tan Tien Commune WTP	30,000	55	Nam Trung Commune WTP	3,500
17	Hong Minh Commune WTP	5,000	**VII**	**Vu Thu District**	
18	Hung Ha Town WTP	10,500	56	Xuan Hoa Commune WTP	10,500
19	Hung Nhan Town WTP	10,000	57	Tu TYan Commune WTP	14,000
IV	**Kien Xuong District**		58	Duy Nhat Commune WTP	3,000
20	Vu Ninh Commune WTP	14,000	59	Viet Hung Commune WTP	1,500
21	Le Loi Commune WTP	9,000	60	Hong Phong Commune WTP	1,500
22	Hoa Binh Commune WTP	2,000	61	Nguyen Xa Commune WTP	5,000
23	Quang Trung Commune WTP	4,000	62	Viet Thuan Commune WTP	5,000
24	Vu Hoa Commune WTP	5,000	63	Minh Lang Commune WTP	9,000
25	Thuong Hien Commune WTP	5,000	64	Bach Thuan Commune WTP	7,000
26	Minh Tan Commune WTP	1,500	65	Hong Ly Commune WTP	2,000
27	Binh Nguyen Commune WTP	10,500	66	Vu Tien Commune WTP	5,000
28	Binh Thanh Commune WTP	9,000	**67**	Vu Thu Commune WTP	30,000
29	Vu Binh Commune WTP	9,000	**VIII**	**Quynh Phu District**	
30	Vu Tay Commune WTP	1,500	68	Quynh Giao Commune WTP	40,000
31	Dinh Phung Commune WTP	1,500	69	An Le Commune WTP	20,500
32	Thanh Ne Town WTP	7,000	70	An Dong Commune WTP	9,500
V	**Thai Thuy District**		71	Quynh Minh Commune WTP	4,500
33	Thuy Lien Commune WTP	10,000	72	An Khe Commune WTP	1,500
34	Thuy Linh Commune WTP	10,500	73	An My Commune WTP	5,000
35	Thai Hoa Commune WTP	15,000	74	An Bai Town WTP	10,000
36	Thai Thinh Commune WTP	10,000	75	Quynh Coi Town WTP	14,500
37	Thai Duong Commune WTP	20,000			

Source : Development plan for water tsupply plant systems for Thai Binh Province (Appedix IX attached to Decision No. 1735/QĐ-TTg dated 29/12/2023 of the Prime Minister for Thai Binh provincial planning for period 2021-2030, vision to 2050)

2.10. Vinh Phuc province

A	WATER SUPPLY	
I	**REHABILITATION, UPGRADING**	
1	Construction, rehabilitation, upgrading water supply system for the rural areas in the province	29
2	Construction, rehabilitation, upgrading water supply system for Son Dong commune, Lap Thach district	1
3	Water supply system for Trung My commune, Binh Xuyen district	1
4	Construction and rehabilitation for Lien Son commune, Lien Hoa from Thai Hoa-Hoa Son WTP	1
5	Construction and upgrading water supply system for Hong Chau commune from Lien Chau-Hong Phuong WTP	1
6	Construction of additional pipelines from rural water works for the whole province	29
7	Upgrading Viet Xuan WTP	1
8	Upgrading Tam Duong WTP	1
9	Upgrading Song Lo WTP (Tam Son)	1
10	Upgrading Ba Hien WTP	1
II	**NEW DEVELOPMENT**	
1	Phuc Binh water treatment plant	1
2	Song Hong water treatment plant	1
3	Don Nhan water treatment plant	1
4	Bo Ly water treatment plant	1

New development water treatment plants in accordance with Report No. 3904/UBND-CN3 dated 26/05/2023 of Vinh Phuc Peoples' Committee

No	Name	Designed capacity of each WTP (m^3/day)	Investment capital (Bil.VND)	Form of investment
1	Phuc Binh Water Treatment Plant	150,000	1,951	Private
2	Red River Water Treatment Plant	29,000	365	Private
3	Water Supply System for Tam Dao and Lap Thach Districts	10,000	194	Private
4	Lo River Water Treatment Plant	16,000	138	Private

Source : Development plan for water treatment plants for Vinh Phuc Province (Appedix XII attached to Decision No. 158/QĐ-TTg dated 06/02/2024 of the Prime Minister on approval of Vinh Phuc provincial planning for period 2021-2030, vision to 2050)

2.11. Quang Ninh province

No.	Name	Quantity of Works
I	**REHABILITATION, UPGRADING**	
1	Zone I:Dong Trieu	3
2	Zone II: Uong Bi City, Quang Yen Town, Hoanh Bo Area (Old) and the western area of Ha Long city	10
3	Zone III: East of Ha Long city, Cam Pha city and Van Don economic zone	8
4	Zone IV: Eastern urban areas (Ba Che, Tien Yen, Dam Ha, Binh Lieu)	6
5	Zone V: Hai Ha town area and Mong Cai city	9
6	Zone VI: Co To island district area	3
II	**NEW DEVELOPMENT**	
6	Zone I:Dong Trieu	2
7	Zone II: Uong Bi City, Quang Yen Town, Hoanh Bo Area (Old) and the western area of Ha Long city	6
8	Zone III: East of Ha Long city, Cam Pha city and Van Don economic zone	4
9	Zone IV: Eastern urban areas (Ba Che, Tien Yen, Dam Ha, Binh Lieu)	13
10	Zone V: Hai Ha town area and Mong Cai city	6
11	Zone VI: Co To island district area	3

Note: For all provinces/cities: The scale, land area usage, and total investment amount for the projects listed above will be calculated, selected, and specifically determined in each stage of investment project preparation, depending on the demand and the ability to balance and mobilize investment capital for each period.

Note: The lacking information/data shall be collected through ATI support or during the site, if selected

Source : Development plan for water treatment plants for Quang Ninh Province
(Appedix XVI attached to Decision No. 80/QĐ-TTg dated 111/02/2023 of the Prime Minister on approval of Quang Ninh provincial planning for period 2021-2030, vision to 2050)

SURVEY RESULT

Table 2B. Promulgated documents on the filed of water supply

No.	Name of provinces/cities	Promulgated documents related to the field of water supply			
		Clean water prices (VND)	Promulgated documents related to the field of water supply	Decision on issuance of water prices	Decision promulgating local technical regulations on quality of clean water used for domestic purposes
1	Ha Noi	Domestic:10m3: 5.973; 10-20m3: 7.052; 20-30m3: 8.669; >30m3: 15.929	Decision No. 69/2013/QD-UBND dated 30/12/ 2013 promulgating regulations on production, supply, use of clean water and protection of water supply works in Hanoi city	Before July 1, 2023, apply Decision 38/2013/QD-UBND, Decision 39/2013/QD-UBND dated September 19, 2013	Being building
		Operating production bussiness and servicw activities: 1. Public administrative agency: 9.955; 2. Public service and public servicw units: 9.955; 3. Material production: 11.615; 4. Service business: 22.068	Decision No. 17/2017/QD-UBND dated 3/5/2017 amending and supplementing a number of articles of regulations on production, supply, use of clean water and protection of water supply works in Hanoi city isued together with Decision No. 69/2013/QD-UBND	- From July 1, 2023, apply Decision No. 3541/QD-UBND dated July 7, 2023.	
			Plan No.148/KH--UBND for safe water supply and prevention off clean water loss and revenue in the Hanoi Area in the period 2017-2020 of the City People's Committee on 01/07/2017		
			The City People's Committee issued Document No. 3473/UBND-DT on water supply zoning and accelerating investment in developing water supply network systems in rural areas on 18/10/2022		
			The Capital's wwater supply planning to 2030, vision to 2050 wwas approved by the Prime Minister in Decision No.499/QD-TTg dated 21/03/2013 and ajusted in Decision No. 554/QD-TTg 6/04/2021		
2	Hai Phong	12,005	Decision No. 487/QD-UBND dated 5/3/2018 of the City People's Committee Approving Hai Phong city water supply planning until 2025, vision to 2050.	Decision No.3208/2016/QD-UBND dated 21/12/2016 of the City People's Committee on clean water prices in Hai Phong city in the period 2017-2019	
			Decision No. 247/2013/QD-UBND dated 29/01/2013 of the City People's Committee promulgating regulations on clean water production, supply and consumption; manage and protect water supply works in Hai Phong.	Decision No.03/2017/QD-UBND dated 14/7/2017 of the City People's Committee on rural clean water prices in Hai Phong city	
3	Hai Duong		Decision No. 2639/QD-UBND dated 24/10/2014 of Hai Duong Provincial People's Committee Master plan for developing clean water supply network of Hai Duong province to 2020, orientation to 2025.	Decision No. 3756/QD-UBND dated 16/12/2016 of Hai Duong Provincial People's Committee on approving the clean water price of Hai Duong Clean Water Trading Company Limited	Decision No. 27/2021/QD-UBND dated 18/11/ 2021 of the People's Committee of Hai Duong province promulgating local technical regulations on clean water quality used for domestic purposes in Hai Duong province

No.	Name of provinces/cities	Promulgated documents related to the field of water supply			
		Clean water prices (VND)	Promulgated documents related to the field of water supply	Decision on issuance of water prices	Decision promulgating local technical regulations on quality of clean water used for domestic purposes
				Decision No. 2539/QD-UBND dated 24/6/2018 of the People's Committee of Hai Duong province on approving the clean water price of Co Dung domestic water service cooperative in Kim Thanh district, Hai Duong province	
4	Nam Dinh		Nam Dinh city water supply planning to 2020, orientation to 2025 was approved by Nam Dinh Provincial People's Committee in Decision No. 605/QD-UBND dated 29/03/2017	Decision No. 56/2021/QD-UBND dated 16/12/ 2021 of Nam Dinh Provincial People's Committee	Being building
			Decision on water supply service area allocation, signing of water supply service agreement: integrated in the investment policy or in the project approval decision	Decision No. 27/2022/QĐ-UBND dated 11/8/2022 of Nam Dinh Provincial People's Committee	
				Decision No. 35/2010/QĐ-UBND dated 27/12/2010 of Nam Dinh Provincial People's Committee	
				Decision No. 06/2019/QĐ-UBND dated 20/3/2019 of Nam Dinh Provincial People's Committee	
				Decision No. 07/2019/QĐ-UBND dated 20/3/2019 of Nam Dinh Provincial People's Committee	
5	Ninh Binh			Decision No. 32/2019/QD-UBND dated 16/09/2019 of Ninh Binh Provincial People's Committee Promulgating clean water consumption tariff in urban areas in Ninh Binh province	Decision No. 26/2022/QD-UBND dated 03/08/2022 of Ninh Binh Provincial People's Committee
				Decision No. 27/2019/QD-UBND dated 28/08/2019 of Ninh Binh Provincial People's Committee regulating the price of clean water consumption in rural areas in Ninh Binh province	
6	Ha Nam	Phu Ly City: 9.620	Decision No. 1055/QD-UBND dated 07/6/2019 of the Provincial People's Committee on approving the Water Supply Plan for Ha Nam province until 2030	Decision No. 59/2022/QD-UBND dated 22/12/2022 of the Provincial People's Committee regulating the selling price of clean domestic water in Ha Nam province in the period of 2023-2024	Decision No. 27/2019/QD-UBND dated 29/08/2019 of the Provincial People's Committee promulgating regulations on coordination in management and monitoring of clean water quality provided to businesses in industrial Zones in Ha Nam province
		Duy Tien Town: 9.000	Decision No. 02/2020/QD-UBND dated February 6, 2020 of the Provincial People's Committee promulgating regulations on management of rural centralized clean water works in Ha Nam province		Decision No. 58/2022/QD-UBND dated 21/12/2022 of the Provincial People's Committee promulgating local technical regulations on clean water quality used for domestic purposes in Ha Nam province
		Kim, Bang District: 9.531			

No.	Name of provinces/cities	Promulgated documents related to the field of water supply			
		Clean water prices (VND)	Promulgated documents related to the field of water supply	Decision on issuance of water prices	Decision promulgating local technical regulations on quality of clean water used for domestic purposes
7	Hung Yen		Decision No. 2416/QD-UBND dated 28/08/2017 of Hung Yen Provincial People's Committee on Clean Water Supply Planning of Hung Yen Province to 2020 and orientation to 2030	Decision No. 814/QD-UBND dated April 15, 2010 of Hung Yen Provincial People's Committee on regulating the selling price of clean water for consumption in Hung Yen province.	
			Decision No. 16/2018/QD-UBND dated 15/04/2018 of the Provincial People's Committee promulgating Regulations on production, supply, use of clean water and protection of centralized water supply works in the Hung Yen province		
8	Bac Ninh		Decision No. 605/QD-UBND dated 27/05/2013 of the People's Committee of Bac Ninh province on approving the regional water supply planning of Bac Ninh province to 2030, vision to 2050	Decision No. 306/QD-UBND dated 14/3/2012 of the People's Committee of Bac Ninh province	Decision 24/2021/QD-UBND dated November 19, 2021 of the People's Committee of Bac Ninh province promulgating local technical regulations on clean water quality used for domestic purposes in Bac Ninh province
			Decision No. 1369/QD-TTg dated 17/10/ 2018 of the Prime Minister on Adjusting the Construction Planning of Bac Ninh province to 2035, vision to 2050	Decision No. 481/QD-UBND dated 18/08/2018 of the People's Committee of Bac Ninh province	
9	Thai Binh	Diem Dien Town: 6.380	Decision No. 1058/QD-UBND of the Provincial People's Committee approving the Water Supply Planning of Thai Binh City until 2020	Decision No. 2024/QD-UBND dated 18/9/2013 of the Provincial People's Committee on approving the clean water consumption price plan of Thai Binh Clean Water JSC	
		Thai Binh City: 7.100		Decision No. 429/QD-UBND dated05/02/2016 of the Provincial People's Committee on approving the clean water consumption price plan of Bitexco Nam Long Joint Stock Company	
		Kien Xuong Town: 7.100		Decision No. 2851/QD-UBND dated 17/10/2016 of the Provincial People's Committee on approving the clean water consumption price plan of Dong Hung Water Supply JSC	
		Vu Quy Urban Area: 7.420		Decision No. 2852/QD-UBND dated 17/10/2016 of the Provincial People's Committee on approving the price plan for domestic clean water consumption of Hoang Dieu Clean Water JSC	
		Quynh Coi Town: 7.000		Decision No. 39/QD-UBND dated 05/01/ 2019 of the Provincial People's Committee on approving the clean water consumption price plan of Thanh Binh Clean Water and Water Industry Supplies JSC	
		An Bai Town: 7.000			

No.	Name of provinces/cities	Promulgated documents related to the field of water supply			
		Clean water prices (VND)	Promulgated documents related to the field of water supply	Decision on issuance of water prices	Decision promulgating local technical regulations on quality of clean water used for domestic purposes
10	Vinh Phuc		Decision No.3878/QD-UBND dated 29/12/2014 of Vinh Phuc PPC on approval of Vinh Phuc urban water supply planning to 2030 and vision to 2050	Decision No. 2755/QD-UBND dated 30/10/2019 of Vinh Phuc PPC on adjusment of domestic water price for rural area in the province	Decision No.43/2019/QD-UBND dated 27/9/2019 of PPC promulgating regulations on management of water supply activities in Vinh Phuc province
11	Quang Ninh				

Note: The lacking information/data shall be collected through ATI support or during the site, if selected

Table 3. Drainage/Sewerage System and Promulgated documents related to the field of drainage/sewerage

Table 3A Wastewater Treatment Plant

Table 3B Development Plan for Wastewater Treatment Plants

Table 3C Promulgated documents related to the field of drainage/sewerage

SURVEY RESULT

Table 3: Drainage/Sewerage System and Promulgated documents related to the field of drainage/sewerage

No	Name of provinces/ cities	Urban level	Separated system			Combined sewer system				Area covered with wastewater collection sytem	Generated wastewater volume	Percentage of household connections
			Total length of pipe/ diameter *km*	Number of wastwater pumping stations/ capacity	Number of drainage pumping station/ capacity	Total length of drainage channel/canal/diameter *km*	Number of regulation lake/ storage area *unit*	Number of wastwater pumping stations/ capacity	Capacity of each wastewater pumping station *m3/s*		*ha*	*%*
1	**Ha Noi**	Special	3,138.05			Sewer: 5183,12km. Main channel: 230,085km	50	59	Main PS: 1. Yen So: 90m3/s 2. Bac Thang Long: 20m3/s; 3. Dong Tru Bridge: 2m3/s, 4. Dong Bong 1: 16m3/s, 5. Dong Bong 2: 12m3/s, 6. Co Nhue 12m3/s, 7. Cau Buou: 5m3/s			20
2	**Hai Phong**	I	6,293			Sewer: 99.165km. Main channel: 259.655km	Regulation lake: 73,2ha	27		1.Vinh Niem: 820ha; 2. Trang Minh: 380ha; 3. Himlam: 12ha		99
3	**Hai Duong**	I										
4	**Nam Dinh**	I	159.68			Channel: 45,61km; Sewer pipe: 271,1	Regulation lake: 41,77ha	2	TT Nam Giang:100.000; Quan Chuot: 57.000m3/h; Kenh Gia: 43.000m3/h			
5	**Ninh Binh**	II										
6	**Thai Binh**	II	30.53	5 PS: PS01: 314,96m3/h; PS02: 250m3/h; PS03: 541,67; PS04833,33m3/h; PS05: 250m3/h		Sewer: 462,317km.Main channel: 150,521km	179/395.09 2,79ha					

No	Name of provinces/ cities	Urban level	Separated system			Combined sewer system				Area covered with wastewater collection sytem	Generated wastewater volume	Percentage of household connections
			Total length of pipe/ diameter *km*	Number of wastwater pumping stations/ capacity	Number of drainage pumping station/ capacity	Total length of drainage channel/canal/diameter *km*	Number of regulation lake/ storage area *unit*	Number of wastwater pumping stations/ capacity	Capacity of each wastewater pumping station *m3/s*		*ha*	*%*
7	**Bac Ninh**	I	Bac Ninh: 292km; Tu Son: Gravity pipe:33km, Pressure pipe: 5,6km	Bac Ninh city: PS1: 203.496m3/year; PS2: 5.960.452m3/year; PS3: 3.455.420m3/year; PS4: 636.285m3/year Tu Son: 11PS			10					
8	**Hung Yen**	II										
9	**Ha Nam**	II	113.2			Sewer: 799,5km; Main channel: 636km	436 /2076,3ha			Bac Chau Giang: 1.490ha, Quy Luu: 180ha, Thanh Chau: 2.400ha, Cau Giat: 235ha, Hoa Mac: 50ha, Dong Van: 150ha, Dai Cuong: 12,4ha; Trung Dong: 18,6ha, Chau river: 3,2ha, Vinh Tru:9,18ha.		
10	**Vinh Phuc**	II	Vinh Yen: Secondary pipeline					3 irrigation PS	Total capacity: 145m3/s Kim Xa: 30m3/s Ngu Kien: 35m3/s Nguyet Duc: 80m3/s		Total: 80146 Urban area: 36974 Rural area: 43172	
11	**Quang Ninh**	I										

Note: The lacking information/data shall be collected through ATI support or during the site, if selected

SURVEY RESULT

Table 3A: Wastewater Treatment Plant

No	Name of provinces /cities	Number of WWTP/ Designed capacity of WWTP/ Stations		Operating capacity of each WWTP/ Station
		Name of WWTP	Designed capacity (m3/day)	
1	Ha Noi	**No. of WWTP: 6**	**276,300**	**203,572**
		Yen So WWTP	200,000	180,000
		Bay Mau lake WWTP	13,300	11,651
		North Thang Long - Van Tri WWTP	42,000	5,921
		Kim Lien wastewater treatment station	3,700	3,700
		Truc Bach wastewater treatment station	2,300	2,300
		Ho Tay WWTP	15,000	
2	Hai Phong	**No. of WWTP: 3**	**38,300**	**30,260**
		Vinh Niem WWTP	36,000	29,000
		Trang Minh WWTP	1,500	1200
		Him Lam WWTP	800	60
		Minh Duc WWTP	1000	
		Tien Lang WWTP	600	
		Cat Ba WWTP	800	
		Tung Dinh WWTP	1500	
3	Hai Duong	**Number of WWTP: 19** ***Total design capacity /47.970m3/day: Urban Area: 21.800m3/day; Industrial Area: 26.170m3/day***	**47,970**	**9,900**
		Ngoc Chau WWTP	13,000	8,000
		WWTP for Hai Duong city ecological tourist and service area	700	700
		WWTP for Thach Khoi residential, trading and handicraft area (stage 1)	350	200
		WWTP for Thai Binh Ecorivers area	5,000	800
		WWTP for new residentail area for Tu Minh ward of Hai Duong city, Tru village of Lai Cach town, Cam Giang district	400	
		WWTP for Western new residential area of Nam Sach Town, Nam Sach district	1,200	
		WWTP for Thanh Cong Eco-urban (stage 1)	350	200
		WWTP for new residentail area for Thanh Quang village-Quoc Tuan district	800	
		WWTP for Nam Sach industrial zone	3,500	
		WWTP for Phuc dien industrial zone	1,500	
		WWTP for Phu Thai industrial zone	500	
		WWTP for Tan Truong industrial zone	4,500	

No	Name of provinces /cities	Number of WWTP/ Designed capacity of WWTP/ Stations		Operating capacity of each WWTP/ Station
		Name of WWTP	Designed capacity (m3/day)	
		WWTP for Dai An industrial zone	2,000	
		WWTP for Extension Dai An industrial zone	2,500	
		WWTP for An Phat High-tech indusrial zone	1,720	
		WWTP for Lai Vu indistructial zone	2,000	
		WWTP for Lai Cach industrial zone	1,000	
		WWTP for Cong Hoa industrial zone	2,000	
		WWTP for Cam Dien-Luong Dien industrial zone	4,950	
4	**Nam Dinh**	Don't have WWTP for domestic have 3 WWTP for 3 industrial zones with total design capacity: 14,500 m3/d operating capacity: 12,500 m^3/d	**4,500**	**3,000**
		WWTP for Hoa Xa Industrial zone	4,500	3,000
5	**Ninh Binh**			
6	**Thai Binh**	**Urban Area**: Thai Binh City WTP; 10.000 m^3/day; **Industry Area:** 1. Phuc Khanh: 2.000 m^3/day; 2. Nguyen Duc Canh: 4.560 m^3/day; 3: Gia Le: 500 m^3/day; 4. Song Tra: 3.700 m^3/day 5. Tien Hai: 3.500 m^3/day; 6. Cau Nghin: 500 m^3/day; 7.Lien Ha Thai: 5.000 m^3/day		**Urban Area**: Thai Binh City WTP; 9.933 m^3/day; **Industry Area**: 1. Phuc Khanh: 792 m^3/day; 2. Nguyen Duc Canh: 776 m^3/day; 3: Gia Le: 350 m^3/day; 4. Song Tra: 353 m^3/day; 5. Tien Hai: 3.500 m^3/day; 6. Cau Nghin: 500 m^3/day; 7.Lien Ha Thai: 110 m^3/day
7	**Bac Ninh**	Bac Ninh: 28.000 m^3/day; Tu Son: 33.000 m^3/day; Phong Khe: 5.000 m^3/day;		Bac Ninh: 18.000 m^3/day; Tu Son: 17.000 m^3/day
8	**Hung Yen**	Number of WWTP: 04/7.620 m^3/day		
9	**Ha Nam**	**Number of WWTP: 13**	**20,475**	**3,450**
		Bac Chau Giang WTP	1,500	-
		Quy Luu WTP	2,500	1,900
		Thanh Chau WTP	5,000	-
		Cau Giat Industrial Cluster WTP	1,200	1,100
		WTP for HM commercial urban area	450	450
		Dong Van Urban Area WWTP	2,230	-
		WWTP for Dai Cuong residential area	250	
		WWTPt in the Middle East area	350	
		WWTP for TEELHOMES Urban Area	215	

No	Name of provinces /cities	Number of WWTP/ Designed capacity of WWTP/ Stations		Operating capacity of each WWTP/ Station
		Name of WWTP	Designed capacity (m3/day)	
		Tan Thanh Town wastewater treatment station No. 1, No. 2	4,200	-
		Kien Khe Town Wastewater Treatment Station No. 01, No. 02	2,000	-
		WWTP for Song Chau Factory in Vinh Tru town	180	-
		WWTP for Residential and commercial factory in Vinh Tru town	400	-
10	Vinh Phuc	Vinh Yen WWTP	5000	
		Vinh Yen WWTP stage 2	6000	
		Investment preparation for Phuc Yen Town WWTP	19300	
		Investment preparation for upgrading Quat Luu WWTP	15000	
		10 wastewater treatment stations for 10 communes	Total capacity : 170 -4 00 m^3/d	
11	Quang Ninh			

Note: The lacking information/data shall be collected through ATI support or during the site, if selected

Table 3B. Development Plan for Wastewater Treatment Plants

(In accordance with the city/provincial development plan to 2030, vision to 2050)

3.1. Hanoi City

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3.2 Hai Phong City

No.	Wastewater drainage Areas	Quantity of proposed WWTPs
I	**Renovate and upgrade wastewater treatment**	
1	Zone 1+2: Central area and Eastern area (including Le Chan, Ngo Quyen, Hong Bang, Kien An, Hai An, Duong Kinh, Do Son and Kien Thuy districts)	3
2	Zone 3: Northern region including Thuy Nguyen district	3
3	Zone 4: Western region including An Duong, An Lao Districts	2
4	Zone 5: Southern region including Tien Lang and Vinh Bao districts	1
5	Region 6: Island area including Cat Hai and Bach Long Vi districts	3
II	**Construction of new wastewater treatment plants**	
1	Zone 1+2: Central area and Eastern area (including Le Chan, Ngo Quyen, Hong Bang, Kien An, Hai An, Duong Kinh, Do Son and Kien Thuy districts)	10
2	Zone 3: Northern region including Thuy Nguyen district	9
3	Zone 4: Western region including An Duong, An Lao Districts	8
4	Zone 5: Southern region including Tien Lang and Vinh Bao districts	10
5	Region 6: Island area including Cat Hai and Bach Long Vi districts	6

Development plan for water supply, sewage networks of Hai Phong City (Appedix VIII attached to Decision No. 1516/QĐ-TTg dated 02/12/2023 of the Prime Minister for Hai Phong provincial planning for period 2021-2030, vision to 2050)

No.	Tên công trình	Location	Planning scale (ha)	Classify
1	Solid waste treatment area in the North of Thuy Nguyen	Thuy Nguyen District	80	Provincial/ inter-district
2	Tran Duong solid waste treatment areas	Vinh Bao District	80	Provincial/ inter-district
3	Trang Cat solid waste treatment areas	Hai An District	60	Provincial/inter-district
4	Dinh Vu solid waste treatment areas	Hai An District	29	Provincial/inter-district
5	Hazardous solid waste treatment areas, district-level waste treatment areas and other waste treatment areas (arranged according to local demands)	Hai Phong City		

Development plan for sewage treatment plants of Hai Phong City (Appedix VIII attached to Decision No. 1516/QĐ-TTg dated 02/12/2023 of the Prime Minister for Hai Phong provincial planning for period 2021-2030, vision to 2050)

3.3 Hai Duong province

No.	Items	Proposed location	Propose capacity (m^3/d)
1	Centralized wastewater treatment plants	Hai Duong city	73,800
2	Centralized wastewater treatment plants	Chi Linh city	43,500
3	Centralized wastewater treatment plants	Kinh Mon town	36,500
4	Centralized wastewater treatment plants	Binh Giang district	25,600
5	Centralized wastewater treatment plants	Gia Loc district	23,500
6	Centralized wastewater treatment plants	Tu Ky district	26,700
7	Centralized wastewater treatment plants	Cam Giang district	30,900
8	Centralized wastewater treatment plants	Nam Sach district	19,100
9	Centralized wastewater treatment plants	Thanh Ha district	20,900
10	Centralized wastewater treatment plants	Kim Thanh district	26,700
11	Centralized wastewater treatment plants	Ninh Giang district	24,800
12	Centralized wastewater treatment plants	Thanh Mien district	20,600
13	Wastewater treatment plants/stations for the residential areas, urban areas and other projectsCác trạm xử lý nước thải của các dự án khu dân cư, khu đô thị, dự án khác	Districts, towns, cities	Conform to the approved projects

Development plan for wastewater treatment plants for Hai Duong Province (Appedix XIII attached to Decision No. 1639/QĐ-TTg dated 19/12/2023 of the Prime Minister for Hai Duong provincial planning for period 2021-2030, vision to 2050)

3.4. Nam Dinh province

No.	Items	Location	Proposed area (ha)
I	**Waste treatment areas**		92
1	Loc Hoa, My Thanh solid waste treatment areas	Nam Dinh City	32.6
2	Industrial and hazaedous solid waste treatment area in Hoa Xa Industrial Park	Nam Dinh City	2.1
3	Lam Town solid waste treatment area	Y Yen	8.5
4	Yen Minh Commune solid waste treatment areas	Y Yen	5
5	Lien Bao commune solid waste treatment areas	Vu Ban	3
6	Nam Giang waste treatment and scrap recycling plants	Nam Truc	3.2
7	Waste treatment complex in the North of Truc Ninh District	Truc Ninh	1.2
8	Nghia Thai Commune solid waste treatment areas	Nghia Hung	6.8
9	Rang Dong Town solid waste treatment areas	Nghia Hung	10
10	Xuan Ninh Commune solid waste treatment areas	Xuan Truong	5
11	Xuan Truong Town solid waste treatment areas	Xuan Truong	1
12	Quat Lam Town solid waste treatment areas	Giao Thuy	13.6
II	**Gathering points/ transfer stations solid waste**	**Districts/Cities**	**57**

Development plan for waste treatment plants for Nam Dinh Province (Appedix X attached to Decision No. 1729/QĐ-TTg dated 29/12/2023 of the Prime Minister for Nam Dinh provincial planning for period 2021-2030)

Table 3B. Development Plan for Wastewater Treatment Plants

3.5. Ninh Binh province

No.	Treatment areas	Location
I	**New Construction**	
2	New construction of 05 wastewater treatment plants in Ninh Binh city	Ninh Binh City
3	New construction of 04 wastewater treatment plants in Tam Diep city	Tam Diep City
4	New construction of 05 wastewater treatment plants in Nho Quan town	Nho Quan Town
5	Me urban areas wastewater treatment plants	Me Town
6	Yen Ninh urban areas wastewater treatment plants	Yen Ninh Town
7	Yen Thinh urban areas wastewater treatment plants	Yen Thinh Town
8	New construction of 02 wastewater treatment plants in Phat Diem urban areas	Phat Diem Town
9	Binh Minh urban areas wastewater treatment plants	Binh Minh Town
10	Wastewater treatment plants in new urban areas and functional areas identified according to urban planning, functional area planning and rural planning	Districts/Cities
II	**Upgrade, Renovation**	
11	Ninh Binh city center wastewater treatment plants	Ninh Binh City

Development plan for wastewater treatment plants for Ninh Binh Province (Appedix XIII attached to Decision No. 218/QĐ-TTg dated 04/03/2024 of the Prime Minister for Ninh Binh provincial planning for period 2021-2030, vision to 2050)

3.6. Ha Nam province

No.	Items	Proposed Location
I	Centralized solid waste treatment plants	
1	The Solid Waste Treatment Plant at Co Chay Valley, Thanh Thuy Commune, Thanh Liem District	Thanh Liem District
2	The Solid Waste Treatment Plant at Dam Gai Valley, Thanh Thuy Commune, Thanh Liem District	Thanh Liem District
3	The But Son, Xuan Thanh, Thanh Thang Cement Plants, etc.	Thanh Liem District, Kim Bang District

Plan for arrangement of waste treatment areas in Ha Nam province (attached to Decision No. 1686/QĐ-TTg dated 26/12/2023 of the Prime Minister for Ha Nam provincial planning for period 2021-2030, vision to 2050)

2.7. Hung Yen province

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3.8. Bac Ninh province

No.	Treatment areas	Location	Proposed area (ha)
1	Northern Solid Waste Treatment Facility (Phu Lang - Que Vo) serving the treatment of domestic solid waste from the central urban area.	Phu Lang Commune	90
2	Plant for Sorting, Recycling and Treatment of Municipal and Industrial Waste at Gia Dong	Thuan Thanh Town	4.5
3	The Waste Treatment Area at Ngu Thai	Thuan Thanh Town	29.3
4	High-tech domestic solid waste treatment plant generates energy in Luong Tai district	Luong Tai District	8.4
5	Planning for Centralized Solid Waste Treatment Area	Yen Phong District, Tu Son city, Tien Du District, Gia Binh District, Bac Ninh City	15ha/area

Development plan for wastewater treatment plants for Bac Ninh Province (Appedix XII attached to Decision No. 1589/QĐ-TTg dated 08 /12/2023 of the Prime Minister for Bac Ninh provincial planning for period 2021-2030, vision to 2050)

3.9. Thai Binh province

No.	Treatment areas	Location	Scope, Scale	Processing technology
1	Centralized solid waste treatment plants	Quynh Phu District	District or inter-district	Advanced, modern, meets legal regulations
2	Centralized solid waste treatment plants	Dong Hung District		
3	Centralized solid waste treatment plants	Thai Thuy District		
4	Centralized solid waste treatment plants (*)	Thai Thuy District		
5	Centralized solid waste treatment plants	Tien hai District		
6	Centralized solid waste treatment plants	Vu Thu District		
7	Centralized solid waste treatment plants	Kien Xuong District		
8	Centralized solid waste treatment plants	Hung Ha District		

Development plan for waste treatment plants for Thai Binh Province (Appedix X attached to Decision No. 1735/QĐ-TTg dated 29/12/2023 of the Prime Minister for Thai Binh provincial planning for period 2021-2030, vision to 2050)

3.10. Vinh Phuc province

No.	Items	Quantity of proposed works
I	**Renovations, Upgrades**	
1	Wastewater treatment plants in Vinh Yen Central areas	1
II	**New Construction**	
1	Western Vinh Yen area wastewater treatment plants	1
2	South Vinh Yen wastewater treatment plants	1
3	Northeast Vinh Yen area wastewater treatment plants	1
4	Phuc Yen wastewater treatment plants	1
5	Wastewater treatment plants in Vinh Yen Central areas	1
6	Drainage and wastewater treatment works of the Water Resources and Flood Management Project in Vinh Phuc province	

Development plan for water supply, sewage systems for Vinh Phuc Province (Appedix XII attached to Decision No. 158/QĐ-TTg dated 06/02/2024 of the Prime Minister for Vinh Phuc provincial planning for period 2021-2030, vision to 2050)

No.	Name of Projects	Proposed Location	Note
1	Khai Quang Ward solid waste treatment areas	Vinh Yen City	New construction
2	Ngoc Thanh commune solid waste treatment areas	Phuc Yen City	New construction
3	Waste recycling and solid waste treatment center in Thanh Cao village, Ngoc Thanh commune	Phuc Yen City	New construction
4	Dao Duc Town solid waste treatment areas	Binh Xuyen District	New construction
5	Solid waste treatment area in the planning proposed area of Son Loi industrial cluster, Son Loi commune	Binh Xuyen District	New construction
6	Upgrading capacity of solid waste treatment area in Hop Hoa town	Tam Duong District	New construction
7	Solid waste treatment areas at the land adjacent to Hop Thanh industrial cluster, Hop Chau town	Tam Dao District	New construction
8	Ngu Kien Ward solid waste treatment areas	Vinh Tuong District	New construction
9	Tho Tang Town and Ta Phu Ward solid waste treatment areas	Vinh Tuong District	New construction
10	Tam Hong Town solid waste treatment areas	Yen Lac District	New construction
11	Nhao Son Ward solid waste treatment areas	Song Lo District	New construction
12	Quang Yen Ward solid waste treatment areas	Song Lo District	New construction
13	Xuan Hoa Ward solid waste treatment areas	Lap Thach District	New construction

Development plan for solid waste treatment areas for Vinh Phuc Province (Appedix XIII attached to Decision No. 158/QĐ-TTg dated 06/02/2024 of the Prime Minister for Vinh Phuc provincial planning for period 2021-2030, vision to 2050)

3.11. Quang Ninh province

Planning of solid waste treatment areas in Quang Ninh province (Appedix XXV attached to Decision No. 80/QĐ-TTg dated 11/02/2023 of the Prime Minister for Quang Ninh provincial planning for period 2021-2030)

No.	Name of Projects	Note
I	**Ha Long City**	
1	Vu Oai and Hoa Binh Communes solid waste treatment center	
2	Solid waste treatment area at Viet Hung Industrial Park, west of the city	New construction
II	**Cam Pha City**	
1	Solid waste treatment area at region 9, Mong Duong ward	New construction
2	Hazardous solid waste treatment area at region 9, Mong Duong Ward, relocated from Quang Hanh	
3	Duong Huy solid waste treatment center, Cam Pha City	
III	**Dong Trieu Town**	
1	Solid waste treatment plant in Truong Luong village, Trang Luong commune, Dong Trieu town	
2	Solid waste treatment area in Tan Tien village, An Sinh commune, Dong Trieu town	
IV	**Uong Bi City**	
1	Khe Giang solid waste treatment area, Thuong Yen Cong commune, Uong Bi city	
2	Lam Thach Cement Factory solid waste treatment area	Co-processing; constructing new waste
V	**Quang Yen Town**	
1	Dong Mai treatment arrea, QuangYen Town	New construction
VI	**Mong Cai City**	
1	Solid waste treatment area km 26 - Quang Nghia commune	
VII	**Hai Ha Commune**	
1	Khu xử lý tại thôn 9, xã Quảng Long,Treatment area at village 9, Quang Long commune, Hai Ha district huyện Hải Hà	New construction
2	Hazardous solid waste treatment area at Texhong Hai Ha Industrial Park	
VIII	**Tien Yen District**	
1	Solid waste treatment area in Dong Hai commune, Tien Yen district	
IX	**Dam Ha District**	
1	Duc Yen Solid waste treatment area	
X	**Van Don District**	
1	Dai Xuyen Solid waste treatment area, van Don district	(Reserved according to general planning)
XI	**Co To District**	
1	Truong Xuan solid waste treatment area	
XII	**Ba Che District**	
1	Khe Ho treatment area	
XIII	**Binh Lieu District**	
1	Vo Ngai Commune treatment area	
2	Hoanh Mo Commune treatment area	

Note: Capacity of the wastewater treatment plants/stations may be adjusted accordingly to thewith actual needs for the projects in conformity with the general urban plan, district regional plan to adapt the social economic development requirements

Note: The lacking information/data shall be collected through ATI support or during the site, if selected

SURVEY RESULT

Table 3C: Promulgated documents related to the field of drainage/sewerage

No	Name of provinces/cities	Promulgated documents related to the field of drainage/sewerage	Regional planning and provincial planning on waste water treatment	Environmental fees and drainage charges
1	Ha Noi		Hanoi Capital Drainage Planning to 2030 and vision to 2050 approved by the Prime Minister in Decision No.725/QD-TTg dated 10/05/2013 The city has built and issued Plan No. 312/KH-UBND dated 28/12/2021 top develop the urban drainage and wwastewater treatment system of Hanoi city Decision 1259/QD-TTG on approval of Ha Noi Capital Master Plan to 2030, vision to 2050; Decision 700/QĐ-TTg on approval of adjusting the Master Plan of Ha Noi Capital to 2045, vision 2065;	
2	Hai Phong	Decision No.47/2017/QDD-UBND dated 29/12/2017 off the City People's Committee promulgating Regulations on management of drainage activities in Hai Phong city	Hai Phong city's wastewater drainage Planningproject to 2025, vision to 2050 was approved by the City People's Committee in Decision No.626/QD-UBNDS dated 27/3/2018 Hai Phong city's elevation and surface drainage Planningproject to 2025, vision to 2050 was approved by the City People's Committee in Decision No.510/QD-UBNDS dated 08/3/2018 Decision 1516/QD-TTG on approval of Hai Phong City Master Plan for period 2021- 2030, vision to 2050;	Environmental fees: 10% Clean water price Drainage charges: 20% Clean water price

No	Name of provinces/cities	Promulgated documents related to the field of drainage/sewerage	Regional planning and provincial planning on waste water treatment	Environmental fees and drainage charges
3	Hai Duong	Hai Duong Provincial People's Committee issued Decision No. 23/2016/QD-UBND dated 05/08/2016 on management of drainage and wastewater treatment activities in Hai Duong province	Hai Duong Provincial People's Committee issued Decision No. 2820/QD-UBND dated 25/11/2013 on approving the strategic orientation for developing a drainage system to collect and treat wastewater in urban areas, craft villages, and industrial zones, industrial clusters and hospitals in Hai Duong province by 2030; estimate costs for implementing Component 2 roadmap and establish orientation.	
		Hai Duong Provincial People's Committee issued Decision No. 1503/QD-UBND dated 10/06/2016 on regulating the collection rate and roadmap for collecting sewerage service fees in Hai Duong city, Hai Duong province.		
			Decision 1639/QD-TTG on approval of Hai Duong Provincial Master Plan for period 2021-2030, vision to 2050;	
4	Nam Dinh	Decision No. 38/2019/QD-UBND dated 24/10/2019 Promulgating regulations on connection to the drainage system in Nam Dinh city of Nam Dinh Provincial People's Committee	Decision No. 1422/QD-TTg dated 17/09/2020 of the Prime Minister on approving the adjustment of Nam Dinh City General Planning to 2040, vision to 2050	
		Decision No. 48/2021/QD-UBND dated October 28, 2021 Promulgating regulations on management of drainage and wastewater treatment activities in Nam Dinh province of Nam Dinh Provincial People's Committee.	Nam Dinh Provincial People's Committee issued Decision No. 2796/QD-UBND dated 10/12/2018 approving the planning project for rainwater and wastewater drainage of Nam Dinh city until 2030, with a vision to 2050.	
			Decision 1729/QD-TTG on approval of Nam Định Provincial Master Plan for period 2021-2030, vision to 2050;	
5	Ninh Binh		Decision 218/QD-TTG on approval of Ninh Bình Provincial Master Plan for period 2021-2030, vision to 2050;	
6	Thai Binh		Decision 1735/QD-TTG on approval of Thái Bình Provincial Master Plan for period 2021-2030, vision to 2050;	Environmental fees: 1. Diem Dien Town: 10% Clean water price; Kien Xuong Town: 10% Clean water price
		Decision No. 02/2016/QD-UBND dated29/01/2016 of Bac Ninh Provincial People's Committee on promulgating regulations on drainage management in Bac Ninh province	Decision 1589/QD-TTG on approval of Bắc Ninh Provincial Master Plan for period 2021-2030, vision to 2050;	

No	Name of provinces/cities	Promulgated documents related to the field of drainage/sewerage	Regional planning and provincial planning on waste water treatment	Environmental fees and drainage charges
7	Bac Ninh	Decision No. 03/2018/QD-UBND dated 31/01/2018 of Bac Ninh Provincial People's Committee on promulgating environmental technical regulations for wastewater applicable in Bac Ninh province		
8	Hung Yen	Decision No. 175/QD-UBND dated 19/01/2023 of Hung Yen Provincial People's Committee approving the Project "Collection and treatment of domestic wastewater in rural residential areas in the province to 2030, with a vision to 2045".		
9	Ha Nam	Ha Nam Provincial People's Committee issued Decision No. 35/2019/QD-UBND dated 9/9/2019 promulgating "Regulations on management of drainage and wastewater treatment activities in Ha Nam province"	Decision 1686/QD-TTG on approval of Hà Nam Provincial Master Plan for period 2021-2030, vision to 2050;	Environmental fees :
				Phu Ly City: 10% Clean water price
				Duy Tien Town: 10% Clean water price
				Kim, Bang District: 10% Clean water price
10	Vinh Phuc	Decision No.21/2018/QD-UBND dated 2/10/2018 of Vinh Phuc PPC promulgating regulations on management of drainage and wastewater treatment activities in Vinh Phuc Province	Decision 158/QD-TTG on approval of Vĩnh Phúc Provincial Master Plan for period 2021-2030, vision to 2050;	
11	Quang Ninh		Decision 80/QD-TTG on approval of Quảng Ninh Provincial Master Plan for period 2021-2030, vision to 2050;	

Note: The lacking information/data shall be collected through ATI support or during the site, if selected

Table 4: Urban drainage and drainage charge in accordance with Decree 80/2015

SURVEY RESULT

Table 4: Urban drainage and drainage charge in accordance with Decree 80/2015

No.	Name of provinces/cities	Urban level	Urban drainage				Drainage service charge in accordance with Decree 80/2015			
			Stormwater drainage catchment	Rainwater storage capacity of lakes/ponds	Quantity and capacity of each urban drainage pumping station	Quantity of flooding points and area of flooding/inundation in the urban	Apllied only environmental protection charge	Being building drainage service charges	Already issued drainage service charges	Applying environmental protection charge and drainage service charge together
			ha	unit	station	point				
1	Ha Noi	Special	31,319	50	59	30 points/ Area of Flooding: 15ha		x		
2	Hai Phong	1		15	8PS: 1. May Den: 32.400m3/h, 2. Vinh niem: 32.400m3/h. 3. Ba Tong: 43.200m3/h; 4. Thuong Ly: 5.400m3/h; 5.South of Cam River:3.600; 6. Trai Chuoi: 3.600; 7. Bach Dang Underpass: 620m3/h; 8. Cau Rao Underpass: 714m3/h	17			x	x
3	Nam Định	II	Nam Dinh City: Northeast Catchment: 640ha, Southwest Catchment: 600ha	Nam Dinh City: 6	Nam Dinh City: 2PS	Nam Dinh City: 8 points/23,5ha				
4	Ha Nam		Phu Ly City: 7.742; Duy Tien Town: 12.091,9; Thanh Liem District: 3.314,8; Ly Nhan District: 1.553,51	436	Phu Ly City: 7PS; Duy Tien Town: 21PS; Thanh Liem District: 5PS; Ly Nhan District: 1PS	Phu Ly City: 7/95ha; Ly Nhan District: 15/74ha	Phu Ly City; Duy Tien Town; Kim Bang District			
5	Thai Binh	II	Thai Binh City: 6.843; Kien Xuong: 1.126,31; Vu Quy: 1.189; Quynh Coi: 90; An Bai: 300	179	18	20.003points/ 3.177,83 ha	Kien Xuong Town; Vu Quy Urban Area	Kien Xuong Town; Vu Quy Urban Area	Thai Binh City; Kien Xuong Town; Vu Quy Urban Area	Kien Xuong Town; Vu Quy Urban Area

Note: The lacking information/data shall be collected through ATI support or during the site, if selected